支援不含ASMLib的ASM資料庫

SnapManager Oracle

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目錄

支援不含ASMLib的ASM資料庫 1
支援的指令碼 1
使用指令碼來支援不含ASMLib的ASM資料庫的限制 1
部署及執行指令碼 2
範例指令碼、可支援不含ASMLib的ASM資料庫 4

支援不含ASMLib的ASM資料庫

根據預設、支援不含ASMLib的ASM。SnapManager基本需求是必須分割用於ASM磁碟群組的裝置。

如果未安裝ASMLib、當您執行下列作業時、與ASM磁碟群組相關的裝置權限會變更為root:disk：

- 重新啟動主機
- 使用Volume Based SnapRestore 功能（VBRSR）從主要儲存設備還原資料庫
- 從次要儲存設備還原資料庫

您可以在SMo.conf中將true指派給oracleasm.support.ino.asmlib組態變數、以設定適當的裝置權限每當從主機新增或移除新裝置時、就會從initasmdisks檔案中新增或移除與ASM磁碟群組相關的裝置。initasmdisks檔案位於/etc/initasmdisks。

例如、如果您設定oracleasm.support.without・asmlib=true、然後執行備份掛載、則會將新裝置新增至initasmdisks。重新啟動主機時、裝置權限和擁有權會由啟動指令碼維護。

> oracleasm.support.without・asmLib的預設值為假。

相關資訊

支援的分割裝置

支援的指令碼

asmmain.sh和asmquerydisk.sh指令碼可讓您變更網格使用者、群組和使用者、所有這些都可用於查詢ASM磁碟。指令碼必須永遠從根執行。

asmmain.sh是從任何新增或刪除裝置的作業中呼叫的主要指令碼檔案。asmmain.sh指令碼會在內部呼叫另一個指令碼、需要從擁有Oracle網格認證的根執行。此指令碼會查詢ASM磁碟群組的裝置、然後在具有裝置權限和擁有權的情況下、將這些項目新增到initasmdisk檔案中。您可以根據環境和僅用於對應/開發/繪圖器/* P1的regex模式、來變更此檔案的權限和擁有權。

asmquerydisk.sh指令碼用於查詢磁碟清單、以建立ASM磁碟群組。您必須根據組態、將值指派給Oracle_base、oracle_homeand oracle_sID。

指令碼位於/opp/NetApp/SMO/plugins/exages/nosmLib。不過、在SnapManager 主機上啟動Oracle伺服器的支援功能之前、必須先將這些指令碼移至/opp/NetApp/SMO/plugins/noasmLib。

使用指令碼來支援不含ASMLib的ASM資料庫的限制

您必須瞭解使用指令碼來支援不含ASMLib的ASM資料庫的特定限制。

- 指令碼可為任何核心版本提供替代解決方案、但僅在未安裝ASMLib時才提供。
- 指令碼的權限必須設定為允許root、GRID、Oracle或同等使用者存取指令碼。

- 指令碼不支援從次要位置還原。

部署及執行指令碼

您可以部署及執行asmmain.sh和asmquerydisk.sh指令碼、以支援不含ASMLib的ASM資料庫。

這些指令碼不遵循指令碼前或指令碼後的語法、啟用intitasmdisks時會呼叫工作流程。您可以在指令碼中變更任何與組態設定相關的項目。建議您執行快速乾燥執行、以確認指令碼中的所有內容是否如預期運作。

> 這些指令碼不會對系統造成故障、也不會影響系統。執行這些指令碼以更新ASM相關磁碟、使其擁有適當的權限和所有權、使磁碟永遠由ASM執行個體控制。

1. 使用分割磁碟建立ASM磁碟群組。
2. 在磁碟群組上建立Oracle資料庫。
3. 停止SnapManager Oracle伺服器的支援。

   > 在RAC環境中、您需要在所有RAC節點上執行此步驟。

4. 修改SMo.conf以包含下列參數：

   a. oracleasm.support.without・asmLib = true
   b. oracleasm.support.without・asmlib.所有權= true
   c. oracleasm.support.without.asmlib.username = ASM執行個體環境的使用者名稱
   d. oracleasm.support.without.asmlib.groupname = ASM執行個體環境的群組名稱

   這些修改只會設定絕對路徑的權限、也就是說、權限只會設定為dm-*裝置、而非分割裝置。

5. 修改在/opp/NetApp/SMO/plugins/exages/nosmLib中可用的外掛程式指令碼、將組態設定納入指令碼中。
6. 在SnapManager 主機上啟動Oracle伺服器的支援功能之前、請先將指令碼複製到/opp/NetApp/SMO/plugins/nosmLib。
7. 瀏覽至/opp/NetApp/SMO目錄、然後執行下列指令碼執行乾燥執行：sh plugins/nosmlib/asmmain.sh

   將會建立etc/initasmdisks檔案、這是使用的主要檔案。

   您可以確認etc/initasmdisks檔案是否包含與已設定的ASM資料庫相關的所有裝置、例如：

```
chown -R grid:oinstall /dev/mapper/360a98000316b61396c3f394645776863p1
         chmod 777 /dev/mapper/360a98000316b61396c3f394645776863p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714239p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714239p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714241p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714241p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714243p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714243p1
```

8. 啟動SnapManager Oracle伺服器的支援功能。
9. 將SnapDrive 下列項目新增至SnapDrive・conffile.disconnect-LUNs-bite-vbsr=on、即可設定適用於UNIX的功能
10. 重新啟動SnapDrive UNIX伺服器的功能。

> 在RAC環境中、您需要針對所有RAC節點執行步驟3至步驟10。

所建立的/etc/initasmdisks檔案、必須從其中一個啟動指令碼或是從rc3.d中新定義的指令碼執行/etc/initasmdisks檔案應一律在oracleha服務啟動之前執行。

範例

```
# ls -ltr *ohasd*
          lrwxrwxrwx 1 root root 17 Aug  7 02:34 S96ohasd ->
/etc/init.d/ohasd
          lrwxrwxrwx 1 root root 17 Aug  7 02:34 K15ohasd ->
/etc/init.d/ohasd
```

在下列範例中、sh -x/etc/initasmdisks預設無法使用、您需要在指令碼中附加它作為函數「state_stack（）」的第一行：

```
start_stack()
{
sh -x /etc/initasmdisks
# see init.ohasd.sbs for a full rationale case $PLATFORM in Linux
}
```

支援不含**ASMLib**的**Oracle RAC ASM**資料庫

如果您使用Oracle RAC資料庫、則只要在主要RAC節點中執行作業、就必須使

用initasmdisks檔案更新RAC節點。

如果不需要驗證即可從主節點登入RAC節點、asmmain.sh會對所有RAC節點執行initasmdisks的安全複製（scp）。每當進行還原時、就會呼叫主要節點的initasmdisks檔案、並可更新asmmain.sh指令碼、以在所有RAC節點中叫用相同的指令碼。

所建立的/etc/initasmdisks檔案、必須從其中一個啟動指令碼或從rcs3. d中新定義的指令碼執行/etc/initasmdisks檔案應一律在oracleha服務啟動之前執行。

支援不含**ASMLib**的**Oracle 10g ASM**資料庫

如果您使用Oracle 10g、則asmcmd命令無法列出磁碟。您可以使用SQL查詢取得磁碟清單。

disk_list.sql指令碼包含在範例目錄中提供的現有指令碼中、以支援SQL查詢。執行theasmquerydisk.sh指令碼時、磁碟list.sSQL指令碼必須手動執行。範例指令碼行會在asmquerydisk.sh檔案中加入附有註解的項目。此檔案可以放在/home/Grid位置、或是您選擇的其他位置。

範例指令碼、可支援不含**ASMLib**的**ASM**資料庫

範例指令碼可在SnapManager Oracle安裝目錄的plugins/examples/nosmlib目錄中找到。

asmmain.sh

```
#!/bin/bash
griduser=grid
gridgroup=oinstall

# Run the script which takes the disklist from the asmcmd
# use appropriate user , here grid user is being used to run
# asmcmd command.
su -c "plugins/noasmlib/asmdiskquery.sh" -s /bin/sh grid
cat /home/grid/disklist

# Construct the final file as .bak file with propre inputs
awk -v guser=$griduser -v ggroup=$gridgroup '/^\/dev\/mapper/ { print
"chown -R "guser":"ggroup" "$1; print "chmod 777 " $1; }'
/home/grid/disklist > /etc/initasmdisks.bak

# move the bak file to the actual file.
mv /etc/initasmdisks.bak /etc/initasmdisks

# Set full full permission for this file to be called while rebooting and
restore
chmod 777 /etc/initasmdisks

# If the /etc/initasmdisks needs to be updated in all the RAC nodes
```

```
# or /etc/initasmdisks script has to be executed in the RAC nodes, then
the following
# section needs to be uncommented and used.
#
# Note: To do scp or running scripts in remote RAC node via ssh, it needs
password less login
# for root user with ssh keys shared between the two nodes.
#
# The following 2 lines are used for updating the file in the RAC nodes:
# scp /etc/initasmdisks root@racnode1:/etc/initasmdisks
# scp /etc/initasmdisks root@racnode2:/etc/initasmdisks
#
# In order to execute the /etc/initasmdisks in other RAC nodes
# The following must be added to the master RAC node /etc/initasmdisks
file
# from the asmmain.sh script itself. The above scp transfer will make sure
# the permissions and mode for the disk list contents are transferred to
the other RAC nodes
# so now appending any command in the /etc/initasmsdisks will be retained
only in the master RAC node.
# The following lines will add entries to the /etc/initasmsdisks file in
master RAC node only. When this script is executed
#  master RAC node, /etc/initasmdisks in all the RAC nodes will be
executed.
# echo 'ssh racnode1 /etc/initasmdisks' >> /etc/initasmdisks
# echo 'ssh racnode2 /etc/initasmdisks' >> /etc/initasmdisks
```

asmquerydisk.sh

```
#!/bin/bash
export  ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/grid/product/11.2.0.3/grid
export ORACLE_SID=+ASM
export  PATH=$ORACLE_HOME/bin:$PATH

# Get the Disk List and save this in a file called dglist.
asmcmd lsdsk  > /home/grid/disklist

# In oracle 10g the above used command 'asmcmd' is not available so use
SQL
# query can be used to take the disk list. Need to uncomment the following
# line and comment the above incase oracle 10g is being in use.
# The disk_list.sql script is availbe in this noasmlib examples folder
itself
# which can be modified as per customer needs.
# sqlplus "/as sysdba" @/home/grid/disk_list.sql > /home/grid/disklist
```

disk_list.sql

```
# su - oracle
-bash-4.1$ cat disk_list.sql
select path from v$asm_disk;
exit
-bash-4.1$
```

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